

# OMNIKON

NATIONAL HACKATHON 2025

ONE MISSION. BUILD THE IMPOSSIBLE.

## TEAM NAME

Duollistes

## MEMBERS

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## THEME

AgriTech / FoodTech

## PROBLEM STATEMENT

Low-cost soil health monitoring software for smallholder farmers. Farmers pick the wrong fertilizers because they lack timely, affordable soil parameter data — private lab tests cost ₹1,200–₹3,000 per sample, and the government Soil Health Card takes 6–8 months to arrive, long after the sowing window. This drives blind, imbalanced fertilizer use (India's actual N:P:K use ratio is ~10.9:4.1:1 against the ideal 4:2:1), which degrades soil health and stagnates yields.

# IDEA TITLE: SoilSaathi

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## SoilSaathi — Smart, Affordable Soil & Fertilizer Advisor

### PROPOSED SOLUTION

**A two-part system:** a low-cost sensor device that reads only the parameters that actually drive fertilizer decisions, paired with an app that converts readings into a crop-specific dosage plan — not just raw numbers.

- Portable device measures pH, N, P, K, Organic Carbon & Moisture — skips the redundant tests bundled into costlier 9-in-1 kits
- App matches readings + crop + season + irrigation type to ICAR/IISS interpretation ranges for an instant dosage recommendation
- Vernacular, icon-first UI so low-literacy farmers can act without help
- Offline-first app; syncs advisory & farm history when connectivity returns
- App contains contacts of agricultural experts which provide real time crop fertilizers recommendation.

### WHY IT'S DIFFERENT

**vs. multi-parameter devices (e.g. ₹16k, 9-in-1 kits):** We test only the parameters proven to drive fertilizer decisions — far lower hardware cost.

**vs. low-cost strips/kits:** Those hand back raw numbers and stop. We close the gap with an actionable dosage plan.

**vs. govt. Soil Health Card:** Results in minutes, not 6–8 months — delivered before the sowing window closes.

# SYSTEM ARCHITECTURE

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## TECHNOLOGY STACK

**Hardware:** pH probe, electrochemical NPK sensor, capacitive soil-moisture sensor, ESP32 microcontroller (BLE/Wi-Fi)

**Mobile/Web App:** React Native (Android-first) + lightweight web dashboard

**Backend:** Python/Node API, cloud DB storing crop, soil-range & farm history data

**Recommendation Engine:** Rule-based ICAR/IISS soil-interpretation logic, refined over time with an ML model as data accumulates

## METHODOLOGY

- Read pH first → flag lime/acidifying need before tuning NPK
- Classify N, P, K as Low / Medium / High vs. ICAR ranges
- Match crop + yield target → build-up, maintenance, or starter-only dose
- Adjust for organic carbon, texture & moisture to finalise the plan

## DATA FLOW

### Soil Sample

Collected in field



### Sensor Device

pH · N · P · K · OC · Moisture



### Mobile App

Bluetooth read-in



### Cloud Engine

Crop DB + ICAR ranges

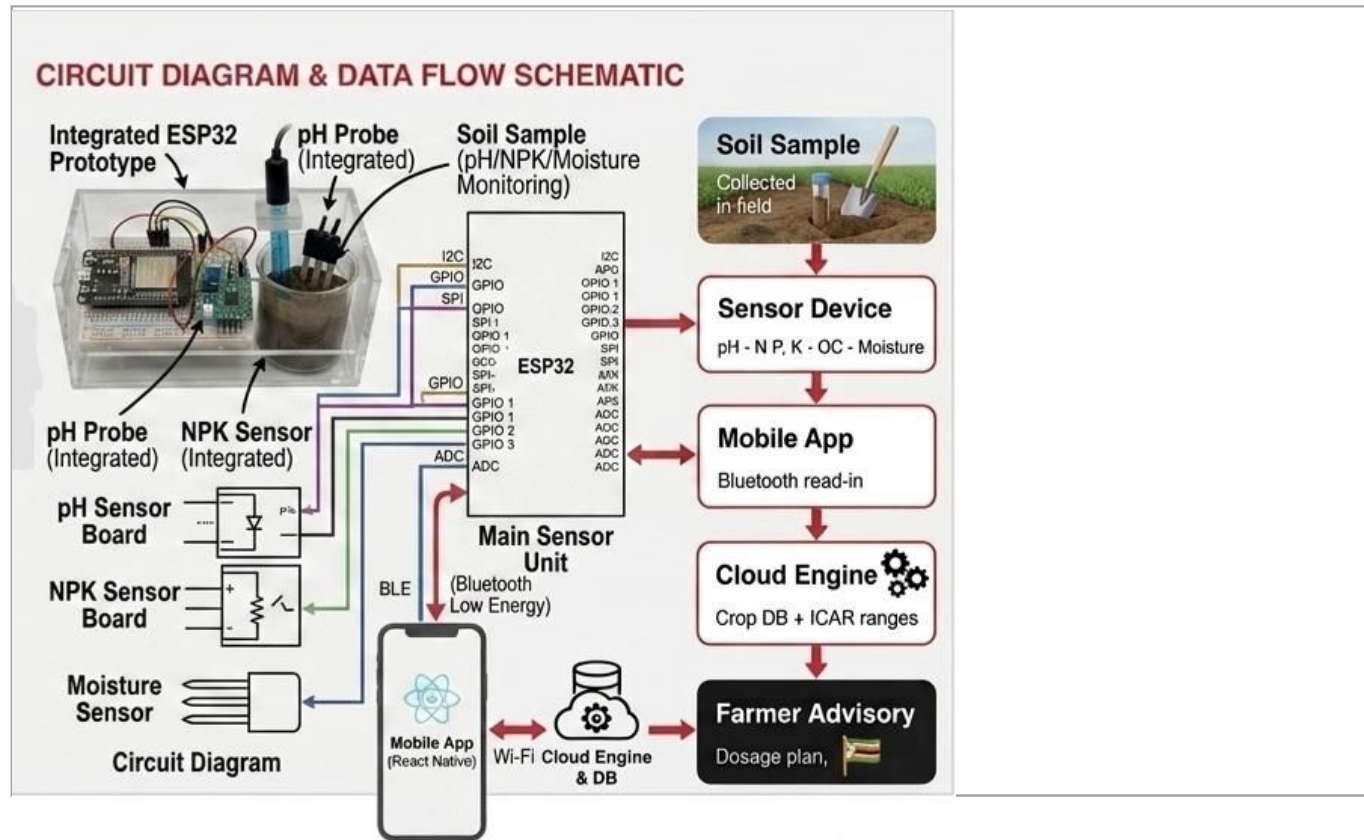


### Farmer Advisory

Dosage plan, local language

# SYSTEM ARCHITECTURE (DESIGN & STRUCTURE)

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## FEASIBILITY

Every input is off-the-shelf: pH/NPK electrochemical sensors and moisture probes are commodity components, the app runs on a standard mobile+cloud stack, and ICAR/IISS soil-interpretation ranges are public data — so the core technical risk is low and the build is achievable within hackathon-to-pilot timelines.

### POTENTIAL CHALLENGES & RISKS

- Low-cost sensor accuracy & drift vs. lab-grade calibration
- Farmer trust — habit and local dealer influence run deep
- Rural connectivity gaps for cloud sync
- Seasonal, spread-out demand complicates distribution

### STRATEGIES TO OVERCOME THEM

- Periodic reference-lab spot checks to auto-correct sensor drift
- Distribute & build trust via FPOs / KVKs / local agri-dealers
- Offline-first app; syncs advisory when signal is available
- Phased rollout — pilot with one FPO cluster before scaling

Reviewer Note:  
FPOs-Farmer Producers Organisations  
KVKs- Krishi Vigyan Kendras

# IMPACT AND BENEFITS

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**Target audience impact:** smallholder farmers get lab-quality soil guidance in minutes instead of 6–8 months, at a fraction of private-lab cost — turning fertilizer spend from guesswork into a targeted investment.

## ECONOMIC

# 60%

*fall in grain output per kg of fertilizer since the 1970s (13–15 kg → 5–6 kg)*

- Balanced K:N dosing can lift rice yields by up to 16%
- Every ₹1 spent on chemical fertilizer today returns just ~88 paise — SoilSaathi targets this waste directly

## ENVIRONMENTAL

# 502

*districts flagged with unsafe groundwater nitrate levels (>45 mg/L)*

- 15–25% of applied nitrogen is lost to leaching & runoff — wasted cost and pollution
- Agriculture drives ~70% of India's N<sub>2</sub>O emissions; precise dosing cuts this at the source

## SOCIAL

# 70%

*of marginal smallholders share a smaller slice of fertilizer-subsidy benefits than large farmers*

- Nitrate-contaminated water raises child health risks (e.g. blue-baby syndrome) in farm households
- Accurate, upfront advisory reduces the input-debt cycle many smallholders fall into

# RESEARCH AND REFERENCES

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## PROBLEM & IDEA RESEARCH

- ICAR — Journal of Soil & Water Conservation ([epubs.icar.org.in](https://epubs.icar.org.in))
- PIB — Official press releases on Soil Health Card scheme
- IFPRI — Understanding Soil Health in Practice: Insights from SHC
- IMPRI — Soil Health Card Scheme: Sustainability review
- Hindustan Times — Making soil testing cheaper, faster, farmer-friendly
- Agrijob — Soil Testing for Farmers: Complete Guide
- <https://epubs.icar.org.in/index.php/JSWC/article/view/175778>
- <https://www.hindustantimes.com/cities/pune-news/making-soil-testing-cheaper-faster-and-farmerfriendly-101763749273037.html>
- <https://www.agrijob.in/soil-testing-farmers-complete-guide/>
- <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2104403&reg=3&lang=2>
- <https://www.ifpri.org/blog/understanding-soil-health-in-practice-insights-from-indias-soil-health-card-scheme/>
- <https://www.impriindia.com/insights/policy-update/soil-health-card-scheme-sustainability/>
- <https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2099759&reg=6&lang=1>

## IMPACT, FEASIBILITY & MARKET

- CNBC-TV18 — Budget 2026: fertilizer allocation & urea subsidy
- IDEAS for India — Balancing fertiliser use for higher yields
- The Week — Unsafe groundwater nitrate levels across 502 districts
- PubMed — Nitrate exposure & health risk indices
- IJSREM — Disparities in irrigation/input subsidy access
- The Better India — NutriSens: IIT Bombay soil-testing device
- <https://www.youtube.com/watch?v=3RllHuyCjZw>
- <https://www.instagram.com/reel/DYVNEOCNLhV/>
- [https://www.linkedin.com/posts/dwivedivivansh\\_agritech-innovation-sustainability-activity-7366055348287262720-NbV3](https://www.linkedin.com/posts/dwivedivivansh_agritech-innovation-sustainability-activity-7366055348287262720-NbV3)
- <https://thebetterindia.com/web-stories/farming/nutrisens-soil-testing-device-iit-bombay-dr-rajul-patkar-farmers-india-11150256>
- <https://pubmed.ncbi.nlm.nih.gov/40956727/>
- <https://visionias.in/current-affairs/upsc-daily-news-summary/article/2025-11-20/the-hindu/environment/hidden-cost-of-polluted-groundwater>
- <https://ijsrem.com/download/disparities-inefficiencies-implications-re-visiting-irrigation-subsidies-for-sustainable-water-use-in-india/>

## REVIEWER NOTES

All statistics are drawn from ICAR/IISS, PIB, CGWB and peer-reviewed sources listed above (2024–2026). Round 1 focuses on idea validation — hardware calibration accuracy will be benchmarked against reference lab results during the prototype phase.